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Data Structures & Algorithms in Python

Stacks and Queues – Tutorials Answers

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Q04

Given the precedence of some operators,

Operators	Precedence
*, /, %	highest
+, -	
<<, >>	
&&	
=	lowest

(a) convert an infix expression, $x = a + b\ c\%d >> e$, to a postfix expression

(b) convert a prefix expression, $= y\&\& << ab >> c + de$, to an infix expression

(c) convert a postfix expression, $xabc\ d\% + e >>=$, to a prefix expression

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Q04 - Answers

(a) convert an infix expression, *x = a + b c%d >> e*, to a postfix expression

Ans: xabc * d% + e >>=

(b) convert a prefix expression, *= y&& << ab >> c + de*, to an infix expression

Ans: y = a << b&&c >> d + e

(c) convert a postfix expression, *xabc d% + e >>=*, to a prefix expression

Ans : = x >> +a% * bcde

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Infix Notation

- While writing an arithmetic expression using **Infix** notation, the operator is placed between the operands.
  - For example, *A+B;* here, plus operator is placed between the two operands A and B.
- **A * (B + C) / D** means: "First add B and C together, then multiply the result by A, then divide by D to give the final answer."
- Information is needed about operator precedence, associativity rules, and brackets which overrides these rules.
- Although it is easy to write expressions using infix notation, computers find it difficult to parse as they need a lot of information to evaluate the expression.

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Postfix Notation

- Postfix notation which is better known as Reverse Polish Notation or RPN.
- In Postfix notation, the operator is placed after the operands. For example, if an expression is written as *A+B* in **Infix** notation, the same expression can be written as *AB+* in **Postfix** notation.
- A postfix operation does not follow the rules of **operator precedence**. The operator which occurs first in the expression is operated first on the operands.
- For example, given a postfix notation **AB+C***. While evaluation, addition will be performed prior to multiplication.
- The order of evaluation of a postfix expression is always from left to right.

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Prefix Notation

- In a **Prefix notation**, the operator is placed before the operands.
- For example, if A+B is an expression in **Infix notation**, then the corresponding expression in prefix notation is given by +AB.
- While evaluating a prefix expression, the operators are applied to the operands that are present immediately on the right of the operator.
- Prefix expressions also do not follow the rules of operator precedence and associativity.
- The expression (A + B) * C is written as: *+ABC in the prefix notation
- So, computers work more efficiently with expressions written using Prefix and Postfix notations.

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Infix to postfix (stack based)

Initialize

- An empty stack for operators.
- An empty list/output for postfix expression.

Scan the infix expression from left to right:

Operand (number/variable): Add it to the postfix output.

**Operator (+, -, , /, ^, ...):*

- While the stack is not empty and the top of the stack has **higher or equal precedence** (and is left-associative), pop from stack to output.
- Push the current operator onto the stack.

Left parenthesis (: Push it on the stack.

Right parenthesis): Pop operators from stack to output until a left parenthesis (is encountered. Remove the (.

After the expression is fully scanned:

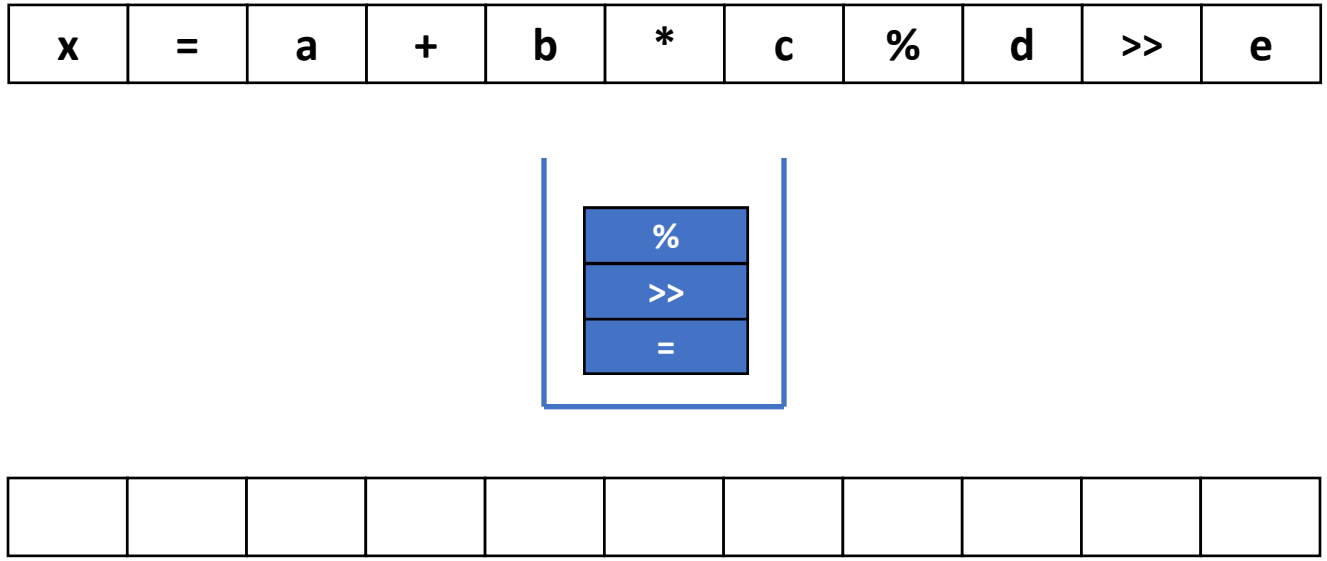
Pop any remaining operators from the stack to the output.

Operators	Precedence
*, /, %	Highest
+, -	
<<, >>	
&&	
=	Lowest

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Q04 – (a) Answer

Convert an **infix** expression, **x = a + b * c%d >> e**, to a **postfix** expression



Operators	Precedence
*, /, %	Highest
+, -	
<<, >>	
&&	
=	Lowest

Infix : An expression is called the Infix expression if the operator appears in between the operands in the expression. Simply of the form (operand1 operator operand2). Example : (A+B) * (C-D)

Postfix: An expression is called the postfix expression if the operator appears in the expression after the operands. Simply of the form (operand1 operand2 operator). **Example : AB+CD-* (Infix : (A+B) * (C-D))**

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Prefix to infix

Do:

- Read the given prefix expression from right to left
- If the character is an operand, push it into the stack.
- But if the character is an operator, pop the top two values from stack.
- Concatenate this operator with these two values *(**1st top value+operator+2nd top value**)* to get a new string.
- Push this resulting string back into the stack.

Until:

- End of prefix expression
- Value in the stack is the desired infix expression.

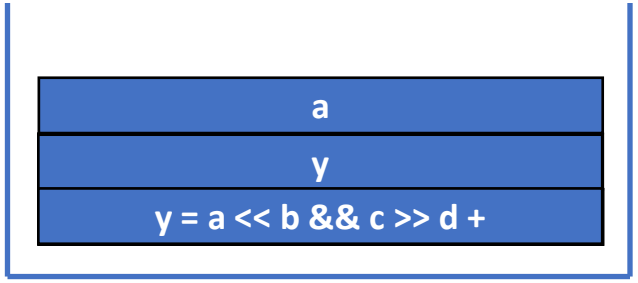
Operators	Precedence
*, /, %	Highest
+, -	
<<, >>	
&&	
=	Lowest

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Q04 – (b) Answer

Convert a **prefix** expression, **= y&& << ab >> c + de**, to an **infix** expression

=	y	&&	<<	a	b	>>	c	+	d	e



y = a << b && c >> d + e

Operators	Precedence
*, /, %	Highest
+, -	
<<, >>	
&&	
=	Lowest

y	=	a	<<	b	&&	c	>>	d	+	e

Prefix : An expression is called the prefix expression if the operator appears in the expression before the operands. Simply of the form (operator operand1 operand2). **Example** : ***+AB-CD (Infix : (A+B) * (C-D))**

Infix : An expression is called the Infix expression if the operator appears in between the operands in the expression. Simply of the form (operand1 operator operand2). **Example** : **(A+B) * (C-D)**

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Postfix to prefix

Do:

- Read the Postfix expression from left to right
- If the symbol is an operand, then push it onto the Stack
- If the symbol is an operator, then pop two operands from the Stack
- Create a string by concatenating the two operands and the operator before them.
  *(string = **operator + operand2 + operand1**)*
- And push the resultant string back to Stack

Until:

- End of prefix expression

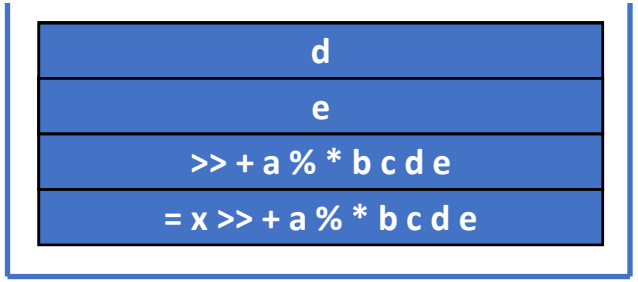
Operators	Precedence
*, /, %	Highest
+, -	
<<, >>	
&&	
=	Lowest

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Q04 – (c) Answer

Convert a **postfix** expression, **xabc * d% + e >>=**, to a **prefix** expression

x	a	b	c	*	d	%	+	e	>>	=



= x >> + a % * b c d e

=	x	>>	+	a	%	*	b	c	d	e

Operators	Precedence
*, /, %	Highest
+, -	
<<, >>	
&&	
=	Lowest

Postfix: An expression is called the postfix expression if the operator appears in the expression after the operands. Simply of the form (operand1 operand2 operator). **Example : AB+CD-* (Infix : (A+B) * (C-D))**

Prefix : An expression is called the prefix expression if the operator appears in the expression before the operands. Simply of the form (operator operand1 operand2). **Example : *+AB-CD (Infix : (A+B) * (C-D))**

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